



Assessment of Serum Cortisol Levels in Patients with Oral Squamous Cell Carcinoma and Its Correlation with Clinical Staging, Histopathological Grading and Nodal Metastasis: A Prospective Study

Veerabhadra Kalyani¹ · Gautam Rao² · Prashanth Hatti³

Received: 26 December 2023 / Accepted: 18 May 2024 / Published online: 1 July 2024
© The Association of Oral and Maxillofacial Surgeons of India 2024

Abstract

Aims and Objectives To assess the serum cortisol level in patients with oral squamous cell carcinoma and correlate this value with clinical staging of tumor using TNM staging, histopathological grading of the tumor using BRYNE'S (1992) invasive tumor front grading system and nodal metastasis using histopathology.

Materials and Methods In this prospective study a total of 25 patients who reported with biopsy proven oral squamous cell carcinoma from Dec 2012–Nov 2014 were included. Patient's clinical parameters were recorded. Clinical staging was assessed using TNM staging. Blood sample was collected from the patient in the early morning and was sent to department of biochemistry, SDM Medical College to assess the serum cortisol levels. The obtained results of serum cortisol levels was correlated with TNM staging, histopathologic grading of the excised tumor (using BRYNE'S grading system) and nodal metastasis (which was confirmed using histopathology of neck specimen). The data was then analyzed statistically.

Results Patients with oral SCC showed morning serum cortisol levels higher. Cortisol levels increased as the stage of the cancer advanced. There was a statistical significance between TNM and cortisol ($p=0.0001$) but no significant correlation between TMS and PN status with cortisol.

Conclusion Patients with advanced stage oral SCC showed significantly higher levels of cortisol than those in an initial stage. This study provides strong evidence that OSSS cells are influenced by neurohormonal mediators and cortisol estimation can be used a biomarker associated with the disease clinical status.

Keywords Serum cortisol level · TNM staging · Brynes histological grading · Oral squamous cell carcinoma

Introduction

Oral cancer represents the third most common form of malignancy in the developing countries [1]. Oral squamous cell carcinoma is the most frequent malignancy in the mouth

✉ Gautam Rao
dr.gautamrao@gmail.com

Veerabhadra Kalyani
kalyaniveerabhadra@gmail.com

¹ Department of Oral Oncology and Reconstruction, Kidwai Peripheral Cancer Center, Kalburgi 585101, India

² Department of Oral and Maxillofacial Surgery, Subbaiah Institute of Medical and Dental Sciences, NH-13, Purle, Shimoga, Karnataka 577222, India

³ Department of Anesthesiology, Kidwai Peripheral Cancer Center, Kalburgi 585101, India

accounting to 95% of all oral malignant lesions [2]. Cancer progression has now been established to be a multifactorial phenomenon.

The main factor which affects tumor progression can be divided into two categories. One is tumor related factor and another is host response to tumor. One of the important factor in the host response to tumor is the lymphocytic infiltration, along the tumor front which provide an immunologic barrier to spread of tumor. This barrier can be affected by the release of cytokine factor released by the tumor which lead to increased cortisol levels in the serum. One of the most important hormone involved in physiological regulation is cortisol, a glucocorticoid produced by the cortex of the adrenal gland. Cortisol secretion occurs in response to physical and psychological stress and its level in the blood are modulated by feedback inhibition mechanism. There are studies which have shown that in tumors of breast, ovary, kidney, lung and colon cancer high cortisol levels have been associated with an advanced clinical stage and they may include immunosuppression and facilitate tumor progression [3]. However this hormonal dysregulation has not yet been investigated extensively in patients with oral cancer. Therefore in this study, we evaluated serum cortisol levels in patients with oral squamous cell carcinoma and correlate it with clinical staging (TNM staging), HPR grading (BRYNE'S grading) and nodal metastasis.

Aims and Objectives

To assess the serum cortisol levels in patients with oral squamous cell carcinoma and correlate this value with.

1. Clinical staging of the tumor using TNM staging
2. Histopathological grading of the tumor using BRYNE'S (1992) invasive tumor front grading system
3. Nodal metastasis confirmed using histopathology

Materials and Methods

This is a prospective study involving 25 biopsy proven cases of squamous cell carcinoma of oral cavity, who were treated in SDM CRANIOFACIAL UNIT between the periods of December 2012–November 2014. This study was approved by institutional review board ethics committee.

Selection Criteria

Inclusion Criteria

- Patients with biopsy proven case of squamous cell carcinoma reporting to CRANIOFACIAL UNIT, SDMCDS, Dharwad.

Exclusion Criteria

- Patient with recurrence of oral cancer.
- Patient with previous radiotherapy.
- Patients with previous history of endocrine disorders.
- Patients who have received steroid therapy in the last 5 years.

Method

Patient's clinical parameters were recorded which include age, sex, tumor location, tobacco and alcohol use and medical history. Through clinical examination in which the site of the lesion and the extent of the lesion were assessed. The clinical staging was assessed by AJCC TNM staging.

Plasma sample

All the patients were fasting in the morning before blood collection, in order to prevent the changes in hormonal levels due to oral intake. Blood samples were collected from the patients in the early morning along with blood for routine blood investigation. Patients' blood was drawn in a plain syringe and sent to department of biochemistry, SDM College of medical sciences and hospital to assess the serum cortisol levels. Then serum cortisol level estimation was done using ADIVA centaur CP assay which is a competitive immunoassay using direct chemiluminescent technology.

The obtained serum cortisol levels was correlated with TNM staging which was done prior to the surgery and then correlated with histopathological grading of the excised tumor using BRYNE'S invasive tumor front grading system and nodal metastasis which was confirmed with histopathology of the neck specimen.

TNM staging by AJCC

**Staging for Cancer of the Oropharynx and Oral Cavity (Including the Lip),
With 5-Year Survival Rates By Stage****Primary Tumor (T)**

TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ
T1	Tumor 2 cm or less in greatest dimension
T2	Tumor more than 2 cm but not more than 4 cm in greatest dimension
T3	Tumor more than 4 cm in greatest dimension
T4	(lip) Tumor invades adjacent structures (eg, through cortical bone, tongue, skin of neck)
T4	(oropharynx and oral cavity) Tumor invades adjacent structures (eg, through cortical bone, soft tissues of neck, deep [extrinsic] muscle of tongue)

Regional Lymph Nodes (N)

NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in a single ipsilateral lymph node, 3 cm or less in greatest dimension
N2	Metastasis in a single ipsilateral lymph node, more than 3 cm but not more than 6 cm in greatest dimension; or in multiple ipsilateral lymph nodes, none more than 6 cm in greatest dimension; or in bilateral or contralateral lymph nodes, none more than 6 cm in greatest dimension
N2a	Metastasis in a single ipsilateral lymph node more than 3 cm but not more than 6 cm in greatest dimension
N2b	Metastasis in multiple ipsilateral lymph nodes, none more than 6 cm in greatest dimension
N2c	Metastasis in bilateral or contralateral lymph nodes, none more than 6 cm in greatest dimension
N3	Metastasis in a lymph node more than 6 cm in greatest dimension

Distant Metastasis (M)

MX	Presence of distant metastasis cannot be assessed
M0	No distant metastasis
M1	Distant metastasis

Stage Grouping		Approximate 5-Year Survival Rate (%)
0	Tis, N0, M0	99%
I	T1, N0, M0	90%–92%
II	T2, N0, M0	75%–85%
III	T3, N0, M0	
	T1, N1, M0	
	T2, N1, M0	50%
IV	T3, N1, M0	
	T4, N0, M0	
	T4, N1, M0	
	Any T, N2, M0	25%–35%
	Any T, N3, M0	
	Any T, Any N, M1	

Adapted with permission from Beahrs OH, Henson DE, Hutter RVP, et al (eds): *Manual for Staging of Cancer*, 4th ed. Chicago, American Joint Committee on Cancer, 1992.

Mor- phologic features	Score			
	1	2	3	4
Degree of kerati- nization (DK)	Highly kerati- nized ($> 50\%$ of the cells)	Moderately kerati- nized ($20\text{--}50\%$ of the cells)	Minimal keratini- zation ($5\text{--}20\%$ of the cells)	No keratiniza- tion ($0\text{--}5\%$ of the cells)
Nuclear polymor- phism (NP)	Little nuclear pleomor- phism ($> 75\%$ mature cells)	Moderately abundant nuclear pleomor- phism		
($50\text{--}75\%$ mature cells)	Abundant nuclear pleomor- phism ($25\text{--}50\%$ mature cells)	Extreme nuclear pleomor- phism ($> 75\%$ mature cells)		
Pattern of invasion (PI)	Pushing, well— delineated infiltrating borders	Infiltrat- ing, solid cords, bands and/ or strands	Small groups or cords of infiltrat- ing cells ($n > 15$)	Marked and widespread cellular dis- sociation in small groups and/or in single cells ($n < 15$)
Lymphop- lasma- cytic infil- tration (LI)	Marked	Moderate	Slight	None
Total malignancy score (TMS)				
4–8 (good)				
9–12 (moderate)				
13–16 (poor)				

Statistical analysis was performed using.

1. One way analysis of variants.
2. Tokay's multiple post-hoc procedure.
3. Spearman's rank correlation or any other suitable tests.

Results

Out of 25 patient, 20 patients (80%) were males and five patients (20%) were females.

25 patients were divided into three age groups. Group 1 from 21 to 40 year which included seven patients (28%),

Group 2 from 41 to 60 years which included 14 patients (56%) and Group 3 above 61 years which included four patients (16%). The mean age of patients was 48.24 years with $SD \pm 12.40$ The mean age of males was 46.95 years with $SD \pm 13.14$ and mean age of females was 53.4 years with $SD \pm 7.77$.

These patients were biopsy proven case of squamous cell carcinoma of oral cavity involving Buccal mucosa(BM) 18 patients (72%), tongue four patients (16%), floor of the mouth(FOM) one patient (4%), Retro mandibular trigone(RMT) and BM one patient (4%), lip and BM 1 patient (4%).

Kruskal Wallis ANOVA test was used to compare the cortisol levels with clinical TNM stage. TNM stage was divided separately according to tumor size (T), regional nodal involvement (N) and distant metastasis (M) and was compared with cortisol levels.

Tumor size was divided into T1, T2, T3 and T4 stage and mean serum cortisol level is calculated in each group. There were no patients in T1 stage. In T2 stage mean serum cortisol level was 9.15 with $SD \pm 1.94$. In T3 stage mean cortisol level was 15.58 with $SD \pm 2.75$ and in T4 stage mean serum cortisol was 20.65 with $SD \pm 1.07$. Cortisol levels was compared with clinical tumor size using Kruskal Wallis ANOVA test. Significant change in cortisol levels were found with the tumor size s, since $p = 0.0001$ i.e. $p < 0.05$. Pair wise comparison by Man-Whitney 'U' test was done within the group which showed a significant change in cortisol levels with tumor size. Since, $p = 0.0062$ i.e. $p < 0.05$.

Regional lymph node was divided into N0, N1, N2 and N3 stages and mean cortisol level was calculated in each group. In N0 mean cortisol level was 9.26 with $SD \pm 2.39$. In N1 mean cortisol level was 14.25 with $SD \pm 4.90$. In N2 stage mean cortisol level was 20.93 with $SD \pm 0.95$ and there were no patients in N3 stage.

Cortisol level was compared with clinical nodal stage using Kruskal Wallis ANOVA test. Significant changes in cortisol levels were found with nodal stage, since $p = 0.0138$ i.e. $p < 0.05$ (Figs. 1, 2, 3, 4, 5, and 6).

Pair wise comparison by Man Whitney 'U' test was done within the group which showed a significant change in cortisol level in between N0 v/s N1 and N0 v/s N2 stage, since $p = 0.033$ i.e. $p < 0.05$. But between stage N1 v/s N2 no much significance was since $p = 0.0807$. In this study there was no incidence of distant metastasis (Tables 1, 2; Figs. 7, 8).

Comparison of overall clinical TNM stage with cortisol levels was done by Kruskal Wallis ANOVA test which showed a significant change in cortisol levels. Since, $p = 0.0001$ i.e. $p < 0.05$ [Table 3, Fig. 9]. Pair wise comparison by Man Whitney 'u' test was done within the group which showed a significant change in cortisol levels. Since, $p = 0.025$ i.e. $p < 0.05$ [Table 3]. But between T2N1M0 v/s



Fig. 1 Patient with carcinoma of buccal mucosa

T2N0M0 and T4N1M0 v/s T4N2M0 there was no much significance. Since, $p=0.999$ [Table 3].

Tumor mean score (TNM) group wise distribution according to BRYNE'S histopathological grading system is divide into three groups. Group1-2 patients (8%), group 2-11 patients (44%) and group-3 12 patients (48%). [Table 4, Fig. 10].

Comparison of TMS score with cortisol levels was done using Kruskal Wallis ANOVA test. In good group mean cortisol level was 14.95 with $SD \pm 9.55$. In moderate group mean cortisol level was 10.96 with $SD \pm 4.35$ and in poor group mean cortisol level was 14.53 with $SD \pm 4.96$. There was no significance change in cortisol levels when compared with TMS score, since $p=0.22$ [Table 5, Fig. 11].

Patients were divided into two two groups after HPR for histopathological nodal status distribution. Group 1 positive node (PN+) consists of 11 patients (44%) and

group 2 negative node (PN-) consists of 14 patients (56%) [Table 6, Fig. 12].

Comparison of PN status with cortisol levels was done using Mann-Whitney test. In PN+ group mean cortisol level was 13.34 with $SD \pm 5.67$ and in PN- group mean cortisol level was 12.71 with $SD \pm 4.87$. There was significant difference in the cortisol levels in both the groups, since $p=0.9346$ [Table 7, Fig. 13].

Correlation between TNM stage, TMS score and PN status with serum cortisol levels was done by Spearman's rank correlation method. Correlation between TNM stage and cortisol level was statistically significant with $p=0.0001$ [Table 8].

There was no statistical significant correlation between TMS and PN status with cortisol levels as $p=0.24$ and 0.93 , respectively [Table 8].



Fig. 2 Plain bulb with blood

Discussion

Oral and oropharyngeal carcinomas are the sixth most common cancer in the world [4]. It is the third most common form of malignancy in the developing countries, while in the developed countries it is the eight most common form of cancer [1]. 57.5% of head and neck cancer occur in Asia. The incidence of oral cancer in India is 30% because of tobacco chewing is used with betel nuts and reverse smoking [5]. Despite evolution in management, the overall survival of patients has not improved significantly during the past 20 years with 5 years survival rate between 45 and 50% [6].

Prognosis of disease depends on various factors like primary tumor, site, stage, tumor thickness, nodal involvement, status of the surgical margins and associated co morbidities. Moreover, cumulative effects of tobacco, betel nut and alcohol decrease the survival rate [7]. The overall national 5 years survival has been reported to vary in range according to tumor size {T1/T2 commonly referred to as “low risk tumor” and T3/T4 commonly referred to as “High risk tumor”}. The outcome is greatly influenced by the stage of the disease especially pathological TNM stage [8].

Studies have shown that cancer progression does not occur only due to local factors related to the tumor, but that it is also influenced by psychoneuroimmunological factors inherent in the patient. The link between psychological and physiological features of cancer risk and progression have been studied through psychoneuroimmunology.

The persistent activation of HPA in the chronic stress response and in depression probably impairs the immune response and contributes to the development and progression of some type of cancer. Stress and depression are associated with decreased cytotoxic T-cells and natural killer cells activities that affect processes such as immune surveillance of tumor and with the events that modulate development and accumulation of somatic mutations and genomic instability [9].

Beside the known effects of the immune system, studies have shown that neurohormonal dysregulation may influence the behavior of certain types of cancer. Here, the influence on tumor microenvironment, viral oncogenesis and immune system is emphasized. There is an evidence linking stress, concomitant behavioral response patterns and resultant neurohormonal and neurotransmitter changes to cancer development and progression. Stress can also influence the

Fig. 3 Blood for centrifugation

expression of viral oncogenes and replication of tumorigenic viruses. Glucocorticoids regulates a wide variety of cellular processes through Glucocorticoids receptor mediated activation or repression of target genes. Glucocorticoids can also activate oncogenic viruses and inhibit antitumor and antiviral cellular immune responses. Glucocorticoids such as cortisol might function in a synergistic fashion with catecholamine's to facilitate cancer growth. In lung carcinoma cells, cortisol increases beta AR density {beta adrenergic receptor} and potentiates the isoproterenol induced increase in cAMP accumulation so it is possible that stressful situations characterized by both increased catecholamine's and cortisol concentrations might have greatest impact on cancer related process [10].

Cancer progression has now been established to be multifactorial phenomenon. The main factors which affects this can be divided into two categories (1) tumor related factor and (2) host response to the tumor.

One of the most important factor in the host response to tumor is the lymphocytic infiltration along the tumor front which provides an immunologic barrier to spread of the tumor. This barrier can be affected by the release of cytokine factor released by the tumor which lead to increased cortisol levels in serum [3].

One of the most important hormone involved in physiological regulation is cortisol, a Glucocorticoids produced by cortex of the adrenal gland in response to ACTH, cortisol secretion occurs in response to physical and psychological

Fig. 4 Serum for cortisol estimation



stress and its levels in blood are modulated by feedback inhibition [6].

In some type of tumors, high cortisol levels have been associated with an advanced clinical stage and they may induce immunosuppression and facilitates tumor progression [11]. Increased cortisol levels have been related to worsen disease prognosis and poorer response to treatment in patients with breast, ovary, kidney, and lung and colon cancer [3, 12–15].

A study was conducted in 31 patients with metastatic breast cancer to know the hormonal levels of HPA system. 15 healthy women were included in the study as control group. The results showed that breast cancer patients had significant elevation in the basal cortisol levels compared to controls. Metastatic breast cancer patients had higher cortisol levels than early stage breast cancer patients. This study concludes that breast cancer is associated with hyperactive adrenal gland, which may be due to physiological

stress associated with presence of metastasis in the circulation [16].

112 patients were examined for the association between IL-6, diurnal cortisol rhythms and feets of depression in the epithelial ovarian cancer patients. Salivary samples were collected three days pre surgery and pre surgical blood sample IL-6 was measured by ELISA and cortisol by a chemiluminiscence immunoassay. Advanced stage ovarian cancer patients demonstrated elevations in cortisol levels. In vegetative and affective depressive symptoms, plasma IL-6 and cortisol when compared with patients of low malignant potential tumors. These results demonstrate significant relationship between IL-6, cortisol and vegetative depression and may have implications for treatment of depression in ovarian cancer patients [14].

Although changes in cortisol levels have been reported in many types of cancers, this hormonal dysregulation has not yet been investigated in patients with oral and oropharyngeal cancer [3].



Fig. 5 ADVIA centaur immunoassay system



Fig. 6 Lite reagent for cortisol estimation

In our study OSCC was more commonly seen in males which is in consensus with the studies done at different centers, which showed that oral cancer is commonest among males in India [7].

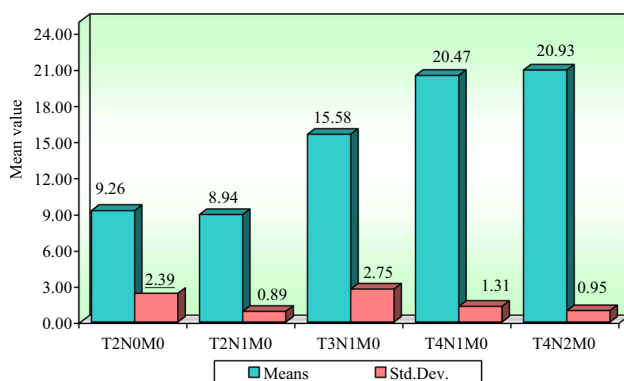
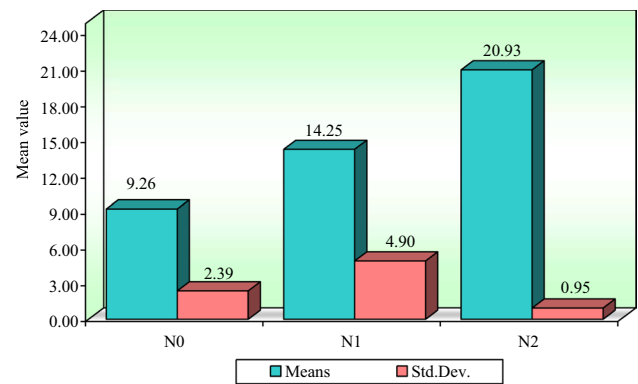
All these patients were biopsy proven case of squamous cell carcinoma of oral cavity involving buccal mucosa, tongue, FOM, RMT and BM, lip and BM. Studies have shown that Oral squamous cell carcinoma may occur anywhere in the mouth, although the most affected sites are

Table 1 Comparison of clinical T stages with cortisol levels (µg/dl) by Kruskal Wallis ANOVA

T stages	Means	Std. dev	Sum of ranks
T2	9.15	1.94	107.00
T3	15.58	2.75	103.00
T4	20.65	1.07	115.00
Total	12.99	5.13	
H-value	18.5785		
p-value	0.0001*		
Pair wise comparison by Mann–Whitney U test			
T2 vs T3	$p=0.0010^*$		
T2 vs T4	$p=0.0012^*$		
T3 vs T4	$p=0.0062^*$		

* $p < 0.05$ **Table 2** Comparison of clinical N stages with cortisol levels (µg/dl) by Kruskal Wallis ANOVA

N stages	Means	Std. dev	Sum of ranks
N0	9.26	2.39	69.50
N1	14.25	4.90	209.50
N2	20.93	0.95	46.00
Total	12.99	5.13	
H-value	8.5610		
p-value	0.0138*		
Pair wise comparison by Mann–Whitney U test			
N0 vs N1	$p=0.0153^*$		
N0 vs N2	$p=0.0339^*$		
N1 vs N2	$p=0.0807$		

* $p < 0.05$ **Fig. 7** Comparison of T stages with cortisol levels (µg/dl)**Fig. 8** Comparison of N stages with cortisol levels (µg/dl)**Table 3** Comparison of TNM stages with cortisol levels (µg/dl) by Kruskal Wallis ANOVA

TNM stages	Means	Std. dev	Sum of ranks
T2N0M0	9.26	2.39	69.5
T2N1M0	8.94	0.89	37.5
T3N1M0	15.58	2.75	103.0
T4N1M0	20.47	1.31	69.0
T4N2M0	20.93	0.95	46.0
Total	12.99	5.13	
H-value	18.5815		
p-value	0.00001*		
Pair wise comparison by Mann–Whitney U test			
T2N0M0 vs T3N1M0	$p=0.0032^*$		
T2N1M0 vs T2N0M0	$p=0.9999$		
T2N1M0 vs T3N1M0	$p=0.0062^*$		
T2N0M0 vs T4N2M0	$p=0.0339^*$		
T2N1M0 vs T4N1M0	$p=0.0254^*$		
T2N0M0 vs T4N1M0	$p=0.0126^*$		
T2N1M0 vs T4N2M0	$p=0.0528$		
T3N1M0 vs T4N1M0	$p=0.0201^*$		
T3N1M0 vs T4N2M0	$p=0.0455^*$		
T4N1M0 vs T4N2M0	$p=0.9999$		

* $p < 0.05$

tongue, lower lip and floor of the mouth. Features of these regions greatly facilitate carcinoma spreading to regional lymph nodes and/or distant organs [6]. Increased levels of

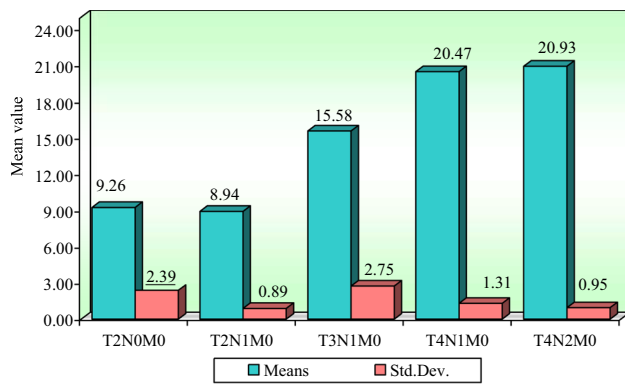


Fig. 9 Comparison of TNM stages with cortisol levels (µg/dl)

Table 4 TMS {tumor mean score} groups wise distribution

TMS	No of patients	% of patients
Good	2	8.00
Moderate	11	44.00
Poor	12	48.00
Total	25	100.00

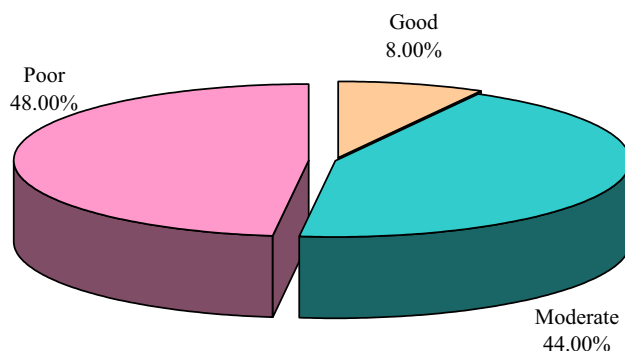


Fig. 10 TMS groups wise distribution

Table 5 Comparison of groups of TMS scores with cortisol levels (µg/dl) by Kruskal Wallis ANOVA

TMS	Mean	Std. dev	Sum of ranks
Good	14.95	9.55	30.50
Moderate	10.96	4.35	111.50
Poor	14.53	4.96	183.00
Total	12.99	5.13	
H-value		2.9749	
p-value		0.2260	

cortisol were most pronounced in oral SCC patients with advanced stage disease (clinical stage 3 and 4). This indicate an HPA axis dysregulation in these patients, associated

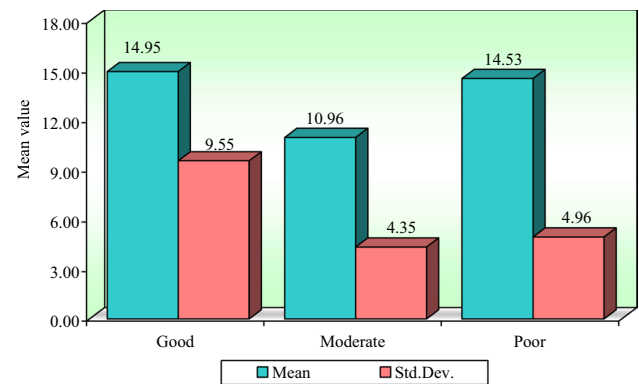


Fig. 11 Comparison of groups of TMS scores with cortisol levels (µg/dl)

Table 6 Histopathological nodal status {PN} distribution of patients

Nodal involvement	No. of patients	Percentage	Cortisol levels	
			Mean	S.D
PN +	11	44	13.3	5.6
PN –	14	56	11.8	5.7

PN –ve 56%	PN +ve 44%

Fig. 12 PN distribution of patients

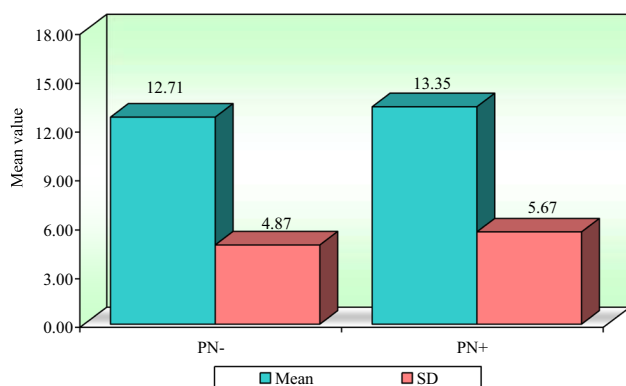
with disease progression. Neuroendocrine changes, such as increased systemic cortisol levels compared with healthy controls, have been observed in patients with tumors in other locations like breast, liver, colon and ovaries [3]. In our study we estimated the cortisol levels in biopsy proven cases of oral squamous cell carcinoma patients and compared with stage of the cancer.

Studies have shown that the risk factors in the healthy volunteers, such as cigarette smoking and alcohol consumption, had no effects on the cortisol levels [3]. Lundstorm et al. did not find difference in morning salivary cortisol levels between drinker and nondrinker in a large sample of volunteers.

In our study clinical staging was assessed by AJCC TNM staging. But Studies have shown that there is a growing concern that TNM staging is insufficient to accurately map or classify OSCC, whose biological impact may be related to volume and pathological aggressiveness of disease. Tumor diameter or surface greatest dimension is used to indicate tumor size in the TNM system [17]. However, this is not the most accurate when compared to tumor thickness or depth of invasion, which can be related directly to prognosis.

Table 7 Comparison of PN status after HPR with cortisol levels ($\mu\text{g/dl}$) by Mann–Whitney U test

PN status after HPR	Mean	SD	Sum of ranks	U -value	Z -value	p -value
PN–	12.71	4.87	180.50	75.50	– 0.0821	0.9346
PN+	13.35	5.67	144.50			

**Fig. 13** Comparison of PN status after HPR with cortisol levels ($\mu\text{g/dl}$)**Table 8** Correlation between TNM stage, TMS and PN– and PN+ with cortisol levels ($\mu\text{g/dl}$) by Spearman's rank correlation method

Variables	Cor	Relation between Cortisol levels ($\mu\text{g}/\text{dl}$) with		
		N	Spearman R	t -value
TNM	25	0.8796	8.8669	0.00001*
TMS	25	0.2412	1.1918	0.2455
PN status after HPR	25	0.0168	0.0804	0.9366

* $p < 0.05$

The main finding of this study was that patients with oral SCC showed higher levels of morning serum cortisol. Cortisol levels increased as the stage of the cancer advanced. So this study provides strong evidence that OSSC cells are influenced by neurohormonal mediators.

Limitations of the Study

1. Sample size was less as the duration of study was two years.
2. There is a need for long term follow up to evaluate the prognosis of the disease and five years disease free survival rate.
3. The analysis of the serum cortisol levels were performed in the samples collected only at one time of day [morning]. It is known that a more careful evaluation of the

HPA is done by identifying the fluctuations of cortisol levels over a daily period

4. We did not evaluate the parameters such as dysphagia, pain, fatigue, sleep dysregulation and psychological stress levels, which could be associated with an increase in cortisol levels

Further studies will be needed to explain the actual role played by dysfunction of the HPA axis in the progression of head and neck cancers.

Conclusion

From this study we conclude that cortisol estimation can be used as a biomarker associated with disease's clinical status.

References

1. Tumuluri VR (1998) A retrospective analysis of cell proliferation in human oral squamous cell carcinoma. Research Institute for Mothers and Infants. The University of Sydney
2. Dantas et al (2003) clinical pathological parameters in squamous cell carcinoma of tongue. *Braz Dent J* 14(1):22–25
3. Bernabe DG, Tamae AC, Miyahara GI, Sundfeld MC, Oliveria SP, Biasoli ER (2012) Increased plasma and salivary cortisol levels in patients with oral cancer and their association with clinical staging. *J Clin Pathol* 65:934–939
4. Warnakulasuriya S (2009) Global epidemiology of oral and oropharyngeal cancer. *Oral Oncol* 45(4):304–316
5. Kenneth D et al (2004) The jaws and oral cavity. *Sternbergs Diagn Surg Pathol* 2:884–915
6. Bagan JV, Scully C (2008) Recent advances in oral oncology 2007 epidemiology, aetiopathogenesis, diagnosis and prognostic. *Oral Oncol* 44(2):103–108
7. Garzino P et al (2006) Clinicopathological parameters and outcomes of 245 patients operated for oral squamous cell carcinoma. *J Craniomaxillofac Surg* 34(6):344–350
8. Noguchi M et al (2003) invasive front in oral squamous cell carcinoma, image and flow cytometric analysis with clinicopathologic correlation. *Oral Surg Oral Med Oral Path Oral Radiol* 93(6):682–687
9. Reiche EM et al (2004) Stress, depression, immune system & cancer. *Lancet Oncol* 5:617–625
10. Atoni MH et al (2006) The influence of biobehavioural factors of tumor biology. *Nat Rev Cancer* 6:240–248
11. Lissoni P et al (2007) Immune and endocrine mechanism of advanced cancer related hypercortisolemia. *Invivo* 21:647–650
12. Lundstrom S, Furst CJ (2003) Symptoms in advanced cancer, relationship to endogenous cortisol levels. *Palliat Med* 17:503–508

13. Rich T, Innominato PF, Boerner J, Mormont MC, Lelobelli S, Baron B, Jasmin C et al (2005) Elevated serum cytokines correlated with altered behavior, serum cortisol rhythm and dampened 24hrs rest activity patterns in patients with metastatic colorectal cancer. *Am Assoc Cancer Res J* 11:1757–1764
14. Lutgendorf SK, Weinrib AZ, Penedo F, Russell D, Degeest K, Costanzo ES et al (2008) interleukin-6, cortisol and depressive symptoms in ovarian cancer patients. *J Clin Oncol* 26(29):4820–4827
15. Bhargava A, Saigal S, Chalishezar M (2010) histopathological grading systems in oral squamous cell carcinoma a review. *J Int Oral Health* 2(4):1–10
16. Pompe GV, Antoni MH, Heijnen CJ (1996) Elevated basal cortisol levels and attenuated ACTH and cortisol response to a behavioral challenge in women with metastatic breast cancer. *J Psychoneuroendocrinol* 21(4):361–374
17. Bernabe DG, Tamae AC, Biasoli ER, Oliveria SH (2011) Stress hormones increase cell proliferation and regulation IL-6 secretion in human oral squamous cell carcinoma cells. *Brain Behav Immun* 25(3):574–583

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.